

A new semiempirical model for the heat and mass transfer inside a spherical catalyst in a stream of hot CH₄/H₂O gases

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HIGHLIGHTS

- Development and validation of a new 0-D model (sub-model) for the steam reforming of methane.
- The new 0-D model includes six gaseous chemical species CH₄, CO₂, CO, H₂O, H₂, N₂.
- Good agreement was achieved between the new 0-D model and the 3D CFD-based model.

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ABSTRACT

This work is devoted to the development and validation of a new 0-D model (sub-model) for the steam reforming of methane inside a spherical catalyst particle (Ni/α-Al₂O₃) placed in a hot methane/steam atmosphere. The particle diameter is 4 mm. The submodel includes six gaseous chemical species (CH₄, CO₂, CO, H₂O, H₂, N₂) and uses experimentally defined reaction rate expressions taken from the literature. The distinguishing feature of the subgrid model is its ability to take into account the internal heat and mass transfer coupled with the heat and mass transfer on the catalyst surface influenced by convection and diffusion around the particle. The sub-model was validated against a comprehensive 3D-CFD-based model resolving the issues of bulk flow, thermal and species boundary layers around the particle and convection and diffusion processes inside the porous catalyst particle. Good agreement was achieved between the new 0-D model and the 3D CFD-based model.

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1. Introduction

Today, the sustainable production of hydrogen is becoming extraordinarily important for the next step of industrial growth combined with a reduction of CO₂ emissions around the world (Dincer and Zamfirescu, 2016). One of the options for the increased production of H₂ is the storage of electricity (during overproduction peaks) in the form of chemicals (Aneke and Wang, 2016; Lu and Nikrityuk, 2018). Steam methane reforming is and will continue to be basically considered a primary technology for the industrial production of hydrogen. For a review of technological developments in hydrogen production, we refer to a review (LeValley et al., 2014) and books (Dincer and Zamfirescu, 2016).

In the design of novel, more effective reactors for steam methane reforming, the important issue is the adequate prediction of the basic characteristics of such devices, e.g., their length and

diameter (see the book (Dincer and Zamfirescu, 2016). Due to the complexity of physical and chemical processes inside reformers, experimental studies are not always capable of characterizing the basic features of all related phenomena. Therefore, computer simulation models have become well-established tools for understanding and optimizing steam-methane reformers, e.g. see the work (Tran et al., 2017; Ngo et al., 2019). It should be noted that all these models use Computational Fluid Dynamics (CFD) equations and algorithms, e.g. see the review (Dixon et al., 2006). For instance, a number of works have been recently published on the numerical modeling of lab-scale reformers, e.g. (Pashchenko, 2018) and industrial-scale reactors (Tran et al., 2017; Ngo et al., 2019; Kuncharam and Dixon, 2020).

In particular, an analysis of these works shows that basically five global models are used:

- an equilibrium model (Pashchenko, 2019; Pashchenko, 2020),
- a CFD-based permeability model (Pashchenko, 2018) (fixed bed is treated as a porous media),

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Nomenclature

A_p	particle surface area	T_p	Particle-averaged temperature inside the particle in 3D model
V_p	particle volume	$Y_{i,in}$	Ambient mass fraction of species i in 0-D model and inflow mass fraction in 3D model
c_p	specific heat capacity	$Y_{i,s}$	Mass fraction of species i on the catalyst surface in 0-D model
D	diffusivity	$Y_{i,p}$	Mass fraction of species i inside the particle in 0-D model
d	diameter	$Y_{i,p}$	Particle-averaged mass fraction of species i inside the particle in 3D model
K	equilibrium constant	Sh	Sherwood number
k_1, k_2, k_3	reaction rate coefficient	Re	Reynolds number
α	stoichiometric coefficient	Sc	Schmidt number
k_{eff}	thermal conductivity	Nu	Nusselt number
ε	particle porosity	Pr	Prandtl number
$\vec{v}$	velocity	ρ	gas density
U_{in}	Inlet gas velocity	ϵ_v	catalyst void fraction
R_g	ideal gas constant	a	adsorption coefficient
h_{pg}	Particle-gas heat transfer coefficient	μ	dynamic viscosity
β_{pg}	Particle-gas mass transfer coefficient	τ	tortuosity
$(hA)_{in}$	Particle internal heat transfer coefficient	P	Pressure
$(\beta A)_{in}$	Particle internal mass transfer coefficient	p_i	Partial pressure of species i
N_p, N_r	number of particles, number of reactions	M	molar mass
$R_{v,r}, R_{s,r}$	volume, surface based reaction rate		
T_{in}	Ambient temperature and inflow temperature in 3D simulations		
T_s	Temperature of the catalyst surface in 0-D model		
T_p	Temperature inside the particle in 0-D model		

- a catalyst-resolved CFD-based model (Dixon, 2017) (transport equations and numerical grid resolve catalyst particles, but particles are treated as porous or solid)
- 1D & 2D mixture-based models (Dixon, 2017; Kuncharam and Dixon, 2020) (heat and mass transfer are modeled using mixture formulation)
- semi-empirical based models (Shigarov, 2020), which use semi-global balance heat and mass transfer equations coupled with closure relations for nondimensional numbers.

It should be critically emphasized that the distinguishing feature of all permeability- or mixture-based CFD simulations of reformers is the use of so-called computational closure relations or equations describing catalyst-gas and catalyst-wall interactions on the single-catalyst-particle level or describing axial and radial dispersion coefficients in fixed beds, respectively, e.g. see the works (Dixon, 2017; Kuncharam and Dixon, 2020).

Recently, with significant progress in multi-scale numerical simulations of dense particulate flows, e.g. see the reviews (Anton et al., 2008; Peng et al., 2020), particle-unresolved Euler-Lagrange models in the form of coupled discrete-element models (DEM) and CFD models are becoming a well-accepted tool for simulations of large-scale particulate systems. In DEM-CFD methods, the fluid phase is treated in Euler space using CFD and the particle trajectories are modeled in Lagrangian space using DEM (Anton et al., 2008; Peng et al., 2020). Depending on the level of coupling, DEMs and CFD models are connected using closure relations (Anton et al., 2008). In particular, to model the heat and mass transfer using a particle-unresolved DEM-CFD model, special closure submodels are required, e.g. see the latest review (Peng et al., 2020). It should be emphasized that particle-unresolved DEM-CFD models showed good capabilities in modeling the heat and mass transfer in complex particulate systems (Bluhm-Drenhaus et al., 2010), where permeability-based CFD models or mixture models fail to predict transport phenomena in particulate media adequately. With this in mind, so-called direct numerical simulations, e.g. see the works (Deen et al., 2014; Chandra et al.,

2020), are used to verify and validate new submodels for mixture-based models or particle unresolved DEM-CFD simulations. One of the first steps in the development and validation of submodels for chemically reacting particles is single particle-resolved CFD, e.g. see the work (Schulze et al., 2013), which can be enhanced to a model for packed beds, e.g. see the work (Schulze and Nikrityuk, 2016).

Referring to catalyst-resolved CFD studies and the development of new submodels for convective-diffusion heat and mass transfer around and inside catalyst, a number of works have been reported in the literature. However, it should be emphasized that pioneering semi-empirical relations for the heat/mass transfer inside the catalyst particles used classical models utilizing pellet effectiveness factor via Thiele module and effective diffusion coefficient of CH_4 , which were derived for steady-state intra-pellet mass/heat transfer in the case of first-order reaction, e.g. see the book (Lapidus and Amundson, 1977). With time progressing, new more complex kinetics, e.g. Langmuir-Hinshelwood-Hougen-Watson (LHHW) kinetics for SMR, were reported in the literature (Hou and Hughes, 2001). Thus, the use of the effectiveness factor for such kinetics, especially for the unsteady intra-pellet mass/heat transfer, is not a trivial task.

Dixon et al. (2010) carried out 3-D CFD-based resolved-particle simulations of a single catalyst particle placed in a stream of hot steam/methane. The commercial software ANSYS-Fluent was used. The catalyst particle was modeled as a porous region and alternatively as a solid, allowing no-slip surface flow boundary conditions on the particle surface. It was demonstrated that using a porous representation of the catalyst particle results in inaccurate temperature and species profiles due to an artifact of convective flux across the particle-fluid interface. It should be noted that this uncertainty was demonstrated for the turbulent flow regime ($Re = 10^4$) using a SST- $k - \omega$ RANS model. We assume that inaccuracies in temperature and species profiles inside the porous catalyst can be attributed to spurious flows inside the porous particle caused by the use of interior boundaries along with the co-located storage of variables. This could be minimized by refining

the mesh inside the particle, which will automatically cause high computational costs. In this view, the use of "solid particle" approach proposed by the authors of (Dixon et al., 2010) is more computationally reliable.

Sulaiman et al. (2019) carried out a numerical investigation using 2D axisymmetric CFD into the interplay between convection, diffusion and reaction, referring to an isothermal solid spherical catalyst particle experiencing a first-order irreversible reaction. The authors additionally established a 0-D model for the mass transfer between the isothermal particle and bulk flow. No any heat transfer model was presented. This model uses a mean volume species concentration to account for diffusion inside the catalyst and internal first-order chemical reactions. The authors employed the decoupled treatment of external and internal mass transfer, where the internal mass transfer coefficient was coupled with kinetics for the first-order chemical reaction. This submodel produced good agreement with isothermal 2D axisymmetric CFD results over a wide range of Reynolds numbers and Schmidt numbers.

In conclusion, it should be noted that recent publications devoted to CFD-based simulations of the heat and mass transfer inside and around a catalyst particle in a hot gas atmosphere, operate with the influence of input parameters such as the particle velocity, ambient temperature or ambient concentration for a single reaction case. However, no efforts have been made to develop and to validate a semiempirical model for multiple reactions cases, such as LHHW kinetics (Hou and Hughes, 2001), referring to steam methane reforming (SMR), which are well-established kinetics used for CFD simulations of SMR (Dixon, 2017; Kuncharam and Dixon, 2020). Such a semiempirical model can serve as a submodel for the heat and mass transfer inside and around a catalyst particle is of great importance in the successful modeling of large-scale reformers using Euler–Lagrange-based models. Additionally, existing isothermal submodels validated against catalyst-resolved CFD operate only with volume-averaged dependent variables, such as the temperature and species concentration, neglecting the heat and mass transfer on the catalyst surface. Motivated by this fact, in this work, we develop a sub-model for the steam reforming of the methane inside a catalyst particle placed in a stream of hot steam/methane atmosphere. This model accounts for the internal heat and mass transfer using surface-averaged and volume-averaged species concentrations instead of only mean volume concentrations. The model is validated against a comprehensive CFD-based model, where the Navier–Stokes equations coupled with the energy and species conservation equations were used to solve the problem by means of the steady state approach.

2. Problem and semi-empirical model formulation

Before we proceed with the formulation of a 0-D model, we next describe the computational domain and input conditions for the model. We consider a porous spherical catalyst particle with a diameter of $d_p = 4$ mm and porosity of $\varepsilon = 0.5$ placed in a hot stream of steam, methane and nitrogen. The inflow Reynolds number $Re = \frac{\rho_{in} U_{in} d_p}{\mu_{in}}$ is 100. The composition of the inflow gas is $Y_{in,CH_4} = 0.3$, $Y_{in,H_2O} = 0.6$ and rest is Y_{in,N_2} . The inflow temperature is varied between 600 K and 1000 K. The ambient pressure is one atmosphere. A diagram of the problem setup is shown in Fig. 1a. In this work we assume that all reactions occur inside the catalyst particle. The choice of Re and the particle size is related to a new fixed bed, see the work (Lu and Nikrityuk, 2018), where the electrical heating is used to drive SMR. Finally, the second reason of small Re is to avoid turbulent flow conditions, which accurate modeling is computationally expensive.

Before we proceed with the set of all equations for our semi-empirical model we introduce a list of assumptions:

- Viscous heating and buoyancy effects are neglected.
- Thermal diffusion is neglected.
- The particle is not-permeable for the gas flow.
- The particle is porous and its porosity is isotropic, the particle porosity is $\varepsilon = 0.5$.

2.1. Mass transfer

The balance of the chemical species on the particle surface takes the following form:

$$\underbrace{\beta_{pg} A_p \rho_{g,s} (Y_{i,in} - Y_{i,s})}_{\text{Convection and Diffusion}} + \underbrace{M_i A_p \sum_r \alpha_{r,i} R_{s,r}}_{\text{Surface reactions}} = \underbrace{(\beta A)_{in} \rho_{g,p} (Y_{i,s} - Y_{i,p})}_{\text{Intraparticle Diffusion}} \quad (1)$$

where β_{pg} is the mass transfer coefficient due to the external convection and diffusion, $(\beta A)_{in}$ is the mass transfer coefficient describing the diffusion inside the particle and $Y_{i,s}$ is the particle surface-averaged mass fraction of species i .

The balance equation for the chemical species inside the particle takes the following form:

$$V_p \varepsilon_p \rho_{g,p} \frac{dY_{i,p}}{dt} = (\beta A)_{in} \rho_{g,p} (Y_{i,s} - Y_{i,p}) + \underbrace{M_i V_p \sum_r \alpha_{r,i} R_{v,r}}_{\text{Reactions inside Particle}} \quad (2)$$

where $Y_{i,p}$ reflects the particle volume-averaged mass fraction of species i . Similar model equations (with different source terms) for the chemical species on the particle surface and for the chemical species inside the particle have been successfully applied for the modeling of char particles gasification in entrained-flow gasifiers. Validations of that methodology against experimental data showed very good agreement, see the work by Schulze et al. (2016). It should be noted that Eq. (2) is written in a general unsteady form. In a steady state, Eq. (2) and Eq. (1) can be combined into one equation:

$$\beta_{pg} A_p \rho_{g,s} (Y_{i,in} - Y_{i,s}) + M_i A_p \sum_r \alpha_{r,i} R_{s,r} + M_i V_p \sum_r \alpha_{r,i} R_{v,r} = 0 \quad (3)$$

Neglecting the term describing chemical reactions on the particle surface, we have the final equation:

$$Y_{i,s} = Y_{i,in} + \frac{M_i V_p \sum_r (\alpha_{r,i} R_{v,r})}{\beta_{pg} A_p \rho_{g,s}} \quad (4)$$

Using that equation, we can calculate $Y_{i,s}$.

After $Y_{i,s}$ is defined, we can calculate $Y_{i,p}$:

$$Y_{i,p} = Y_{i,s} + \frac{M_i V_p \sum_r (\alpha_{r,i} R_{v,r})}{(\beta A)_{in} \rho_{g,p}} \quad (5)$$

Using Eqs. (4) and (5) we finally have:

$$Y_{i,p} = Y_{i,in} + M_i V_p \sum_r (\alpha_{r,i} R_{v,r}) \left(\frac{1}{\beta_{pg} A_p \rho_{g,s}} + \frac{1}{(\beta A)_{in} \rho_{g,p}} \right) \quad (6)$$

Finally, we emphasize that Eq. (5) is fully implicit equations because terms which include $R_{v,r}$ depends nonlinearly on $Y_{i,p}$. However, the form of Eq. (5) corresponds to the fixed-point numerical method, which was used to solve this equation for each species. To illustrate the nonlinearity of Eq. (5) we write this equation in the full form for $Y_{CH_4,p}$:

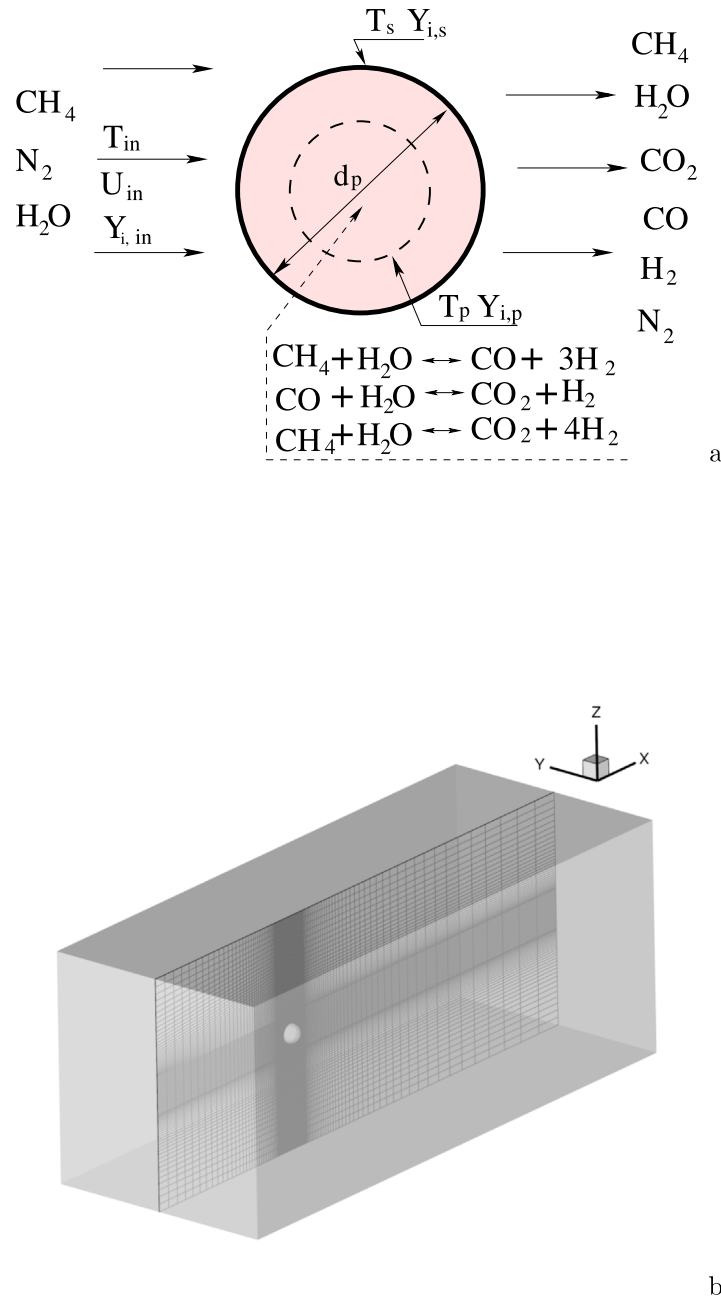


Fig. 1. Diagram illustrating the catalyst particle places in a stream of hot gas consisting of methane and steam (a), 3D domain ($30d_p \times 12d_p \times 12d_p$) and grid ($2.2 \cdot 10^6$ CV, where $0.5 \cdot 10^6$ CV were used inside the sphere) for CFD simulations (b).

$$Y_{CH_4,p} = Y_{CH_4,s} + \frac{M_{CH_4} V_p}{(\beta A)_{in} \rho_{g,p}} \left(-\frac{k_1 \left(p_{CH_4} p_{H_2O}^{0.5} - p_{H_2}^3 p_{CO} / K_1 p_{H_2O}^{0.5} \right)}{p_{H_2}^{1.25} \text{den}^2} - \frac{k_3 \left(p_{CH_4} p_{H_2O} - p_{H_2}^4 p_{CO_2} / K_3 p_{H_2O} \right)}{p_{H_2}^{3.5} \text{den}^2} \right) \quad (7)$$

where $p_{CH_4} = \frac{Y_{CH_4} P M}{M_{CH_4}}$. For mathematical expressions of terms in that equation we refer to Section 2.3 and Section 2.4.

We want to emphasize that the chemical reactions on the particle surface were neglected because the ratio between the particle surface and overall particle surface participating in the reactions is $6 \cdot 10^{-3}\%$ for $d_p = 4$ mm. The surface area (BET, m^2/g) of the catalyst is $14.3 \text{ m}^2/\text{g}$, see the work (Hou and Hughes, 2001).

2.2. Heat transfer

An equation describing the heat balance on the catalyst surface takes the following form:

$$\underbrace{(hA)_{in}(T_s - T_p)}_{\text{Internal Diffusion}} = \underbrace{h_{pg} A_p (T_{in} - T_s)}_{\text{External Convection and Diffusion}} + \underbrace{\epsilon \sigma A_p (T_{in}^4 - T_s^4)}_{\text{Radiation}} + \underbrace{A_p \sum_r^{N_r} (-\Delta H_r) R_{s,r}}_{\text{Surface reactions}} \quad (8)$$

where T_s is the particle surface-averaged temperature. The emissivity of the particle surface was set to 0.7. It should be noted that T_{in} corresponds to the inlet temperature of the gas and at the same time in the problem we solve it equals to the ambient temperature of the gas T_∞ . For dense bed systems (fixed beds or fluidized beds)

T_{in} will correspond to the gas phase temperature inside a control volume where the particle is placed. The following works (Schulze and Nikrityuk, 2016; Lu and Nikrityuk, 2018) show formulations of Eq. (8) for cases with multiple particles. The radiative heat transfer between the particle surface and the bulk flow having the temperature $T = T_{in} = T_{\infty}$ is modeled using the Stefan-Boltzmann term $\epsilon \sigma A_p (T_{in}^4 - T_s^4)$, which is well acceptable approximation for the modeling of the radiative heat transfer between a solid surface and bulk gas flow. For an example we refer to the works by Schulze et al. (Schulze et al., 2016) (single particle case) and by Lu and Nikrityuk (Lu and Nikrityuk, 2018) (multiple particles case).

The particle temperature T_p can be calculated similarly to two temperature 0-D model developed for an ice-particle melted in a stream of hot water (Bansal and Nikrityuk, 2017):

$$m_p c_p \frac{dT_p}{dt} = (hA)_{in} (T_s - T_p) + \underbrace{V_p \sum_r^{N_r} (-\Delta H_r) R_{V,r}}_{\text{Reactions inside Particle}} \quad (9)$$

where T_p reflects the volume-averaged temperature of the particle. It should be noted that similar two temperature 0-D model has been validated against experimental data referring to phase change phenomena, see the works (Schmidt et al., 2014; Bansal and Nikrityuk, 2017; Bansal et al., 2017). In a steady state, both equations can be combined into one equation:

$$h_{pg} A_p (T_{in} - T_s) + \epsilon \sigma A_p (T_{in}^4 - T_s^4) - A_p \sum_r^{N_r} (-\Delta H_r) R_{s,r} + V_p \sum_r^{N_r} (-\Delta H_r) R_{V,r} = 0 \quad (10)$$

Neglecting the impact which chemical reactions on the particle surface exert on the overall heat rate, we may write the final equation:

$$h_{pg} A_p (T_{in} - T_s) + \epsilon \sigma A_p (T_{in}^4 - T_s^4) + V_p \sum_r^{N_r} (-\Delta H_r) R_{V,r} = 0 \quad (11)$$

That equation can be easily solved using the Newton method to find T_s .

Finally, in a steady state, the particle temperature T_p can be predicted using the following equation:

$$T_p = T_s + \frac{V_p \sum_r^{N_r} (-\Delta H_r) R_{V,r}}{(hA)_{in}} \quad (12)$$

The system of equations comprising of Eqs. (4), (6), (11) and (12) is solved implicitly via Matlab. All equations, except Eq. (10), whose solution was found applying the Newton method, were calculated utilizing a fixed-point method with underrelaxation. The iterations were stopped when the estimated relative error for each dependent variable ($T_s, T_p, Y_{i,s}, Y_{i,p}$) was less than 10^{-6} .

2.3. Closure relations

The external mass transfer coefficient β_{pg} is calculated from the Sherwood number correlation.

$$Sh = \frac{\beta_{pg} \cdot 2r_p}{D_i} \quad (13)$$

Likewise, the heat transfer coefficient is calculated from the Nusselt number.

$$Nu = \frac{h_{pg} \cdot 2r_p}{k_g} \quad (14)$$

The Sherwood and Nusselt numbers are obtained using the Whitaker correlation (Whitaker, 1972):

$$Nu = 2 + \left(0.4Re^{0.5} + 0.06Re^{0.67}\right) Pr^{0.4} \left(\frac{\mu}{\mu_s}\right)^{0.25} \quad (15)$$

In this work we assume that the Lewis number equals unity ($Le = 1$), which is reasonable assumption for hot gases (Schulze et al., 2013; Schulze et al., 2016), thus, we may write the following $Nu = Sh$:

$$Sh = 2 + \left(0.4Re^{0.5} + 0.06Re^{0.67}\right) Sc^{0.4} \left(\frac{\mu}{\mu_s}\right)^{0.25} \quad (16)$$

This statement can be approved by the fact that if the Lewis number equals to unity then $Pr = Sc$.

The particle Reynolds number is calculated using

$$Re_p = \frac{\rho_{in} \mathbf{U}_{in} \cdot 2r_p}{\mu_{in}} \quad (17)$$

The diffusion coefficient D_i is obtained by using the Lewis number equals to 1.

$$Le = \frac{k_g}{D_i c_p \rho} = 1; \quad D_i = \frac{k_g}{c_p \rho} \quad (18)$$

where transport properties k_g, c_p and μ are functions of T and are calculated utilizing polynomials taken from the data base (Bride et al., 1993).

The mass and heat transfer coefficients for the intraparticle mass and heat transfer are defined as follows (Schmidt et al., 2014):

$$(\beta A)_{in} = 4\pi D_{eff} \left(\frac{1}{a_1 r_p} - \frac{1}{r_p}\right)^{-1} \quad (19)$$

$$(hA)_{in} = 4\pi k_{eff} \left(\frac{1}{a_1 r_p} - \frac{1}{r_p}\right)^{-1} \quad (20)$$

where $a_1 = 0.85$ (Schmidt et al., 2014)

The effective diffusion coefficient inside the porous particle D_{eff} is expressed using void volume fraction and tortuosity.

$$D_{eff} = \frac{\epsilon}{\tau} D_i \approx \epsilon^2 D_i \quad (21)$$

It should be noted that this relation corresponds to micropore-macropore model of Wakao and Smith (Wakao and Smith, 1962). In this work we use this relation only for purposes of validation against 3D CFD. For comparisons with experimental data we have to use exact values of τ measured in experiments. Plus, the Knudsen diffusion should be taken into account if we want to compare our model with experiments.

2.4. Kinetics

Reaction rates $R_{V,r}$ for each chemical reaction are calculated from reaction kinetics (Hou and Hughes, 2001), where the reaction kinetics for steam reforming are given in the units of $[\text{mol}/\text{m}^3 \cdot \text{s}]$.



$$R_{V,1} = \frac{k_1 (p_{\text{CH}_4} p_{\text{H}_2\text{O}}^{0.5} - p_{\text{H}_2}^3 p_{\text{CO}} / K_1 p_{\text{H}_2\text{O}}^{0.5})}{p_{\text{H}_2}^{1.25} \text{den}^2} \quad (23)$$



$$R_{V,2} = \frac{k_2 (p_{\text{CO}} p_{\text{H}_2\text{O}}^{0.5} - p_{\text{H}_2} p_{\text{CO}_2} / K_2 p_{\text{H}_2\text{O}}^{0.5})}{p_{\text{H}_2}^{1.25} \text{den}^2} \quad (25)$$



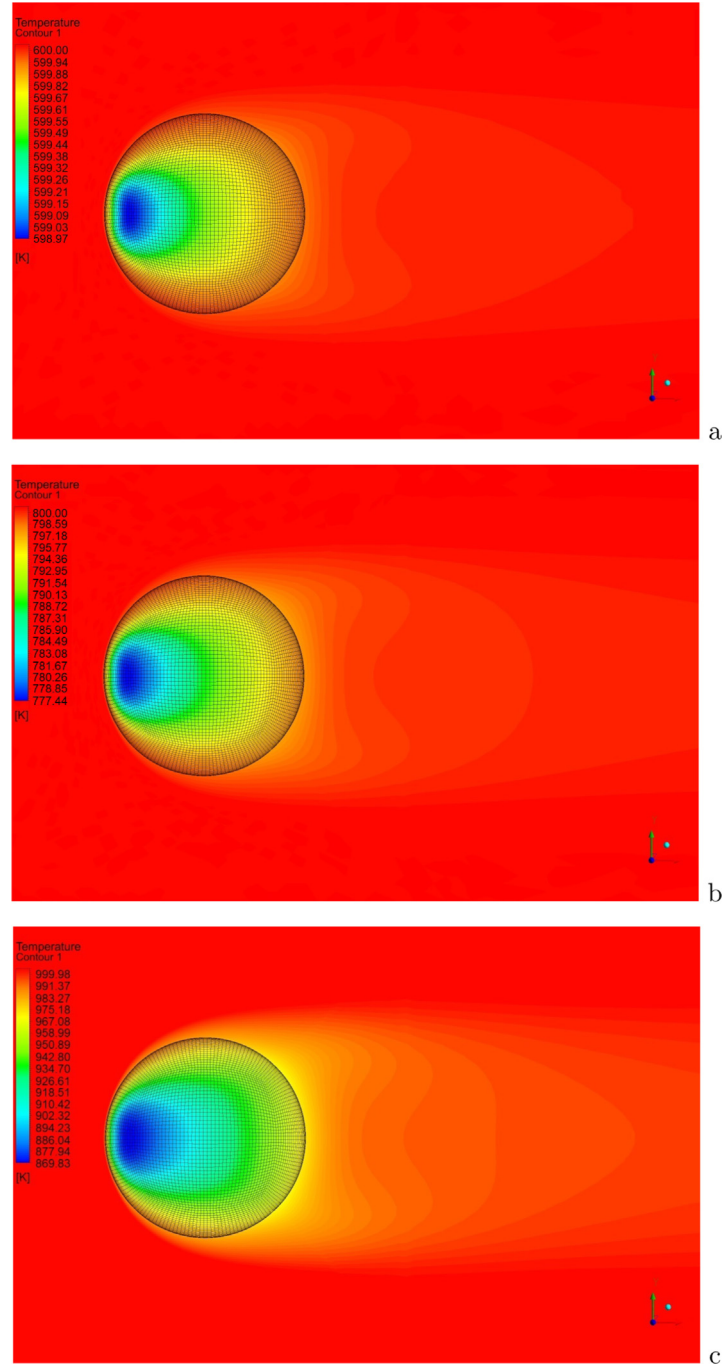


Fig. 2. Contour plots of the temperature on a slice through the midplane of the particle (YX plane, for $z = 0$) predicted for $Re = 100$ and different ambient temperatures: a - $T_{in} = 600K$, b - $T_{in} = 800K$, c - $T_{in} = 1000K$. The flow is from left to right.

$$R_{V,3} = \frac{k_3 (p_{CH_4} p_{H_2O} - p_{H_2}^4 p_{CO_2} / K_3 p_{H_2O})}{p_{H_2}^{3.5} \text{den}^2} \quad (27)$$

$$\text{den} = 1 + K_{CO} * p_{CO} + K_H * p_{H_2}^{0.5} + K_{H_2O} * p_{H_2O} / p_{H_2} \quad (28)$$

According to the work (Hou and Hughes, 2001), the rate coefficients are

$$k_1 = 5.922 \cdot 10^8 \exp\left(\frac{-209200}{R_g T}\right) \left[\text{kmol/s/kg/kPa}^{0.25}\right] \quad (29)$$

$$k_2 = 6.028 \cdot 10^{-4} \exp\left(\frac{-15400}{R_g T}\right) \left[\text{kmol/s/kg/kPa}\right] \quad (30)$$

$$k_3 = 1.093 \cdot 10^3 \exp\left(\frac{-109400}{R_g T}\right) \left[\text{kmol/s/kg/kPa}^{0.25}\right] \quad (31)$$

The adsorption coefficients take the following forms (Hou and Hughes, 2001):

$$K_{H_2O} = 9.251 \cdot \exp\left(\frac{-15900}{R_g T}\right); K_H = 5.68 \cdot 10^{-10} \exp\left(\frac{93400}{R_g T}\right) \left[\text{kPa}^{-0.5}\right] \quad (32)$$

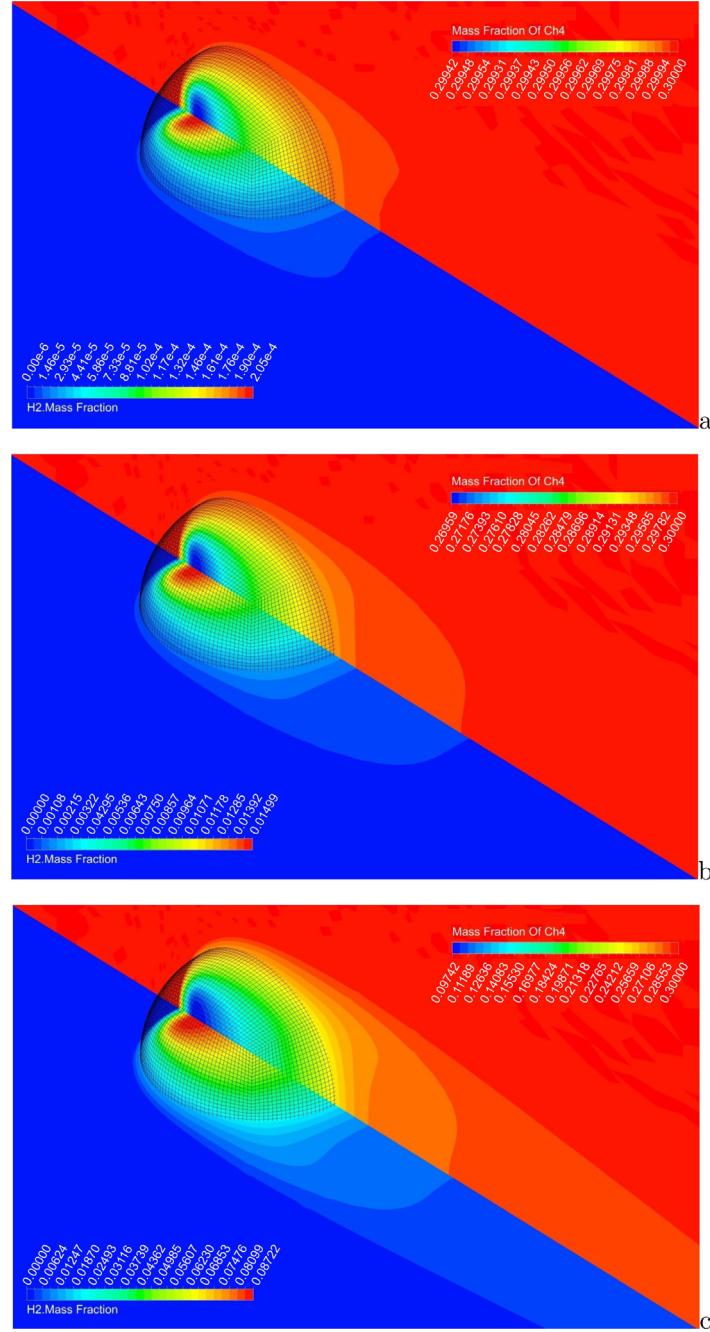


Fig. 3. 3D distribution of CH_4 and H_2 mass fractions inside the particle predicted for $Re = 100$ and different ambient temperatures: a - $T_{in} = 600\text{K}$, b - $T_{in} = 800\text{K}$, c - $T_{in} = 1000\text{K}$.

$$K_{\text{CO}} = 5.127 \times 10^{-13} \exp\left(\frac{140000}{R_g T}\right) [\text{kPa}^{-1}] \quad (33)$$

In this work we used the following expressions for the equilibrium coefficients:

$$K_1 = 1.198 \cdot 10^{17} \exp\left(\frac{-26830}{T}\right) [\text{kPa}^2]; K_2 = 1.767 \cdot 10^{-2} \exp\left(\frac{4400}{T}\right) [\text{kPa}] \quad (34)$$

$$K_3 = 2.117 \cdot 10^{15} \exp\left(\frac{-22430}{T}\right) [\text{kPa}^2] \quad (35)$$

Finally, source terms in Eq. (2) and the subsequent equations take the following form for each specific species:

$$\text{CH}_4 : \sum_r^{N_r} (\alpha_{r,i} R_{V,r}) = -(R_{V,1} + R_{V,3}) \quad (36)$$

$$\text{H}_2\text{O} : \sum_r^{N_r} (\alpha_{r,i} R_{V,r}) = -(R_{V,1} + R_{V,2} + 2R_{V,3}) \quad (37)$$

$$\text{CO} : \sum_r^{N_r} (\alpha_{r,i} R_{V,r}) = R_{V,1} - R_{V,2} \quad (38)$$

$$\text{CO}_2 : \sum_r^{N_r} (\alpha_{r,i} R_{V,r}) = R_{V,2} + R_{V,3} \quad (39)$$

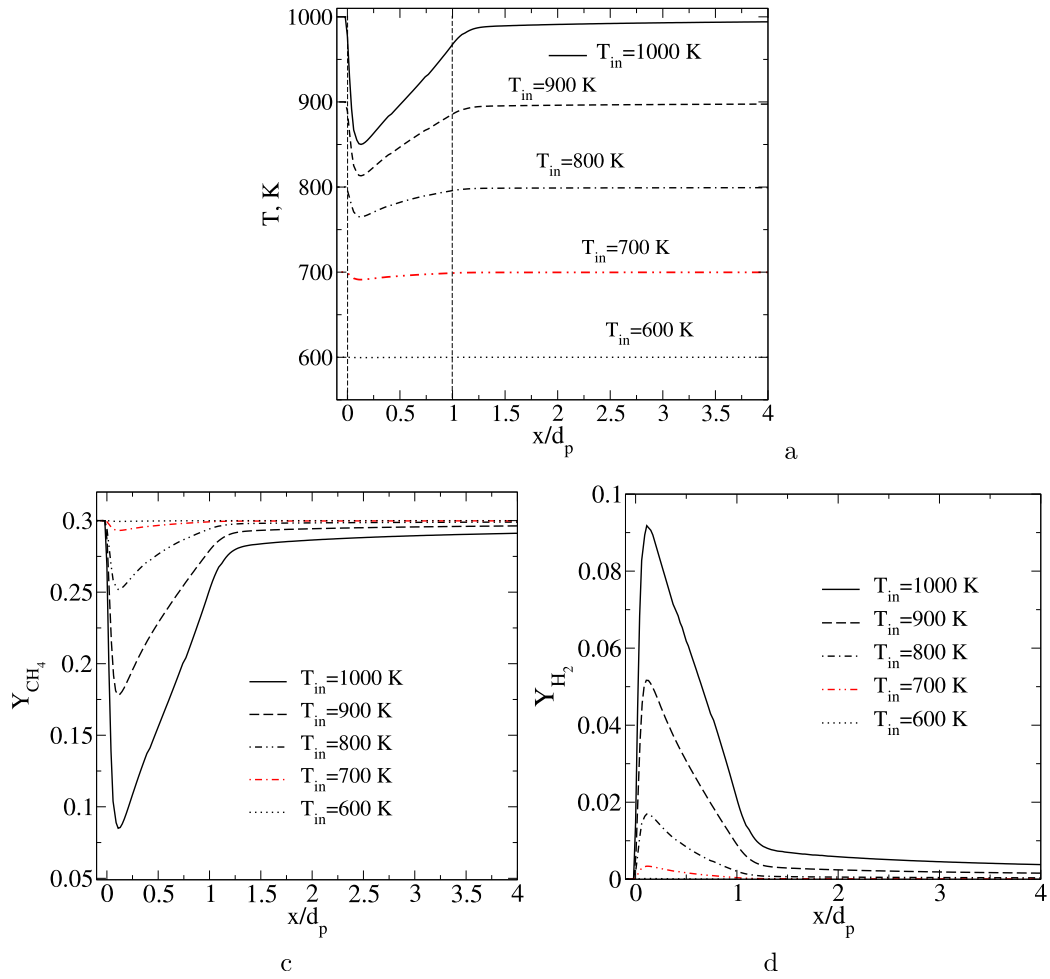


Fig. 4. Axial profiles of the temperature, T (a), methane mass fraction, Y_{CH_4} (b) and hydrogen mass fraction, Y_{H_2} along the particle symmetry axis at $z = 0$ and $y = 0$ predicted using 3D CFD for different values of the ambient temperature T_{in} and constant $Re = 100$.

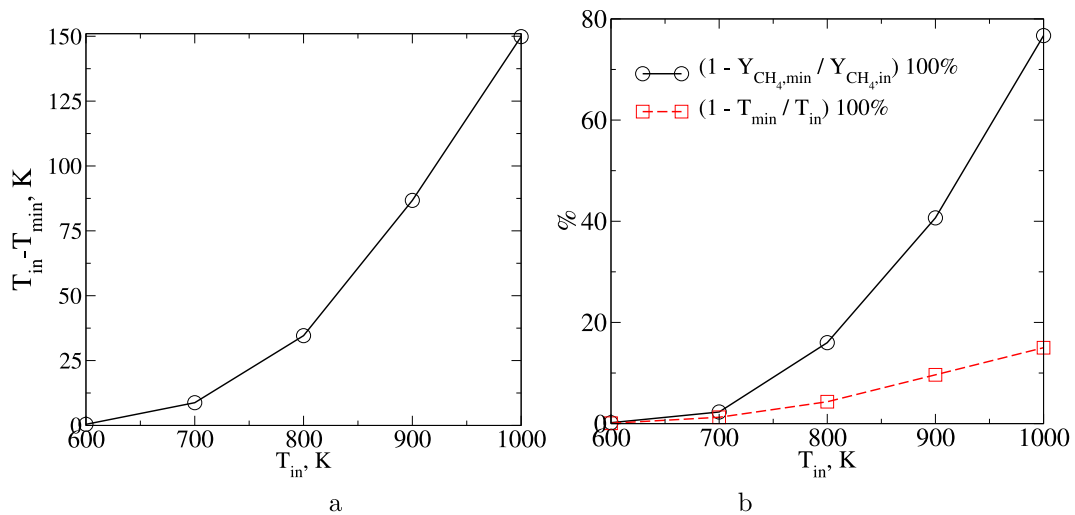


Fig. 5. Change of temperature ($T_{in} - T_{min}$) (a) and relative differences in the temperature $\frac{T_{in} - T_{min}}{T_{in}}$ and CH_4 mass fraction (b) between ambient values and minimum values inside the particle depending on the ambient temperature T_{in} .

$$H_2 : \sum_r^{N_r} (\alpha_{r,i} R_{V,r}) = 3R_{V,1} + R_{V,2} + 4R_{V,3} \quad (40)$$

The reaction enthalpy for reactions Eq. (22), Eq. (24) and Eq. (26) are $-\Delta H_1 = -206.1$ kJ/mol, $-\Delta H_2 = 41.2$ kJ/mol and $-\Delta H_3 = -165.0$ kJ/mol, respectively. As a reference we used the works (Dixon,

2017; Pashchenko and Eremin, 2021). The relation between the partial pressure and mass fraction is

$$p_i = \frac{Y_i PM}{M_i}; \quad M = \frac{1}{\sum_i^N \frac{Y_i}{M_i}} \quad (41)$$

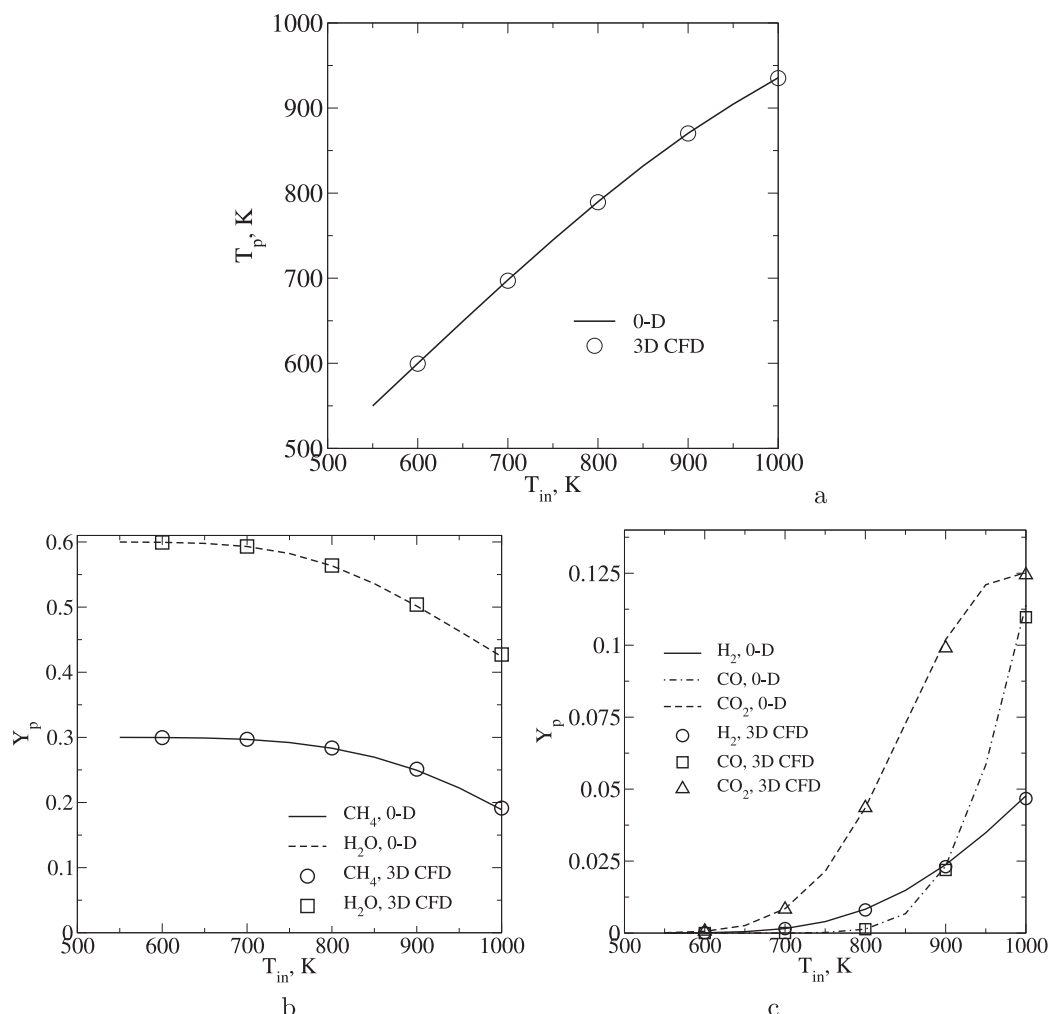


Fig. 6. 3D CFD vs 0-D: comparison of the particle-averaged temperature (Eq. (46)) against T_p calculated using Eq. (12) (a) and particle-averaged mass fractions of species (Eq. (47)) (b), (c) obtained from 3D CFD simulations and T_p , $Y_{i,p}$ predicted using the 0-D model (Eq. (6)) for $Re = 100$.

3. Validation against 3D CFD

To validate our semi-empirical model we use 3D CFD simulations as a validation tool. A diagram of the computational domain is given in Fig. 1b. The size of the 3D domain corresponds to $30d_p \times 12d_p \times 12d_p$ in x, y and z directions, respectively. The flow field is directly determined by the inflow, with the mean velocity having a constant value across the inflow plane. The temperature at the inflow exhibits a constant, uniform profile. Far away from the particle, in directions normal to the free stream, we assume, that no more velocity or temperature gradients are present, so all surrounding domain boundaries excluding the in- and outflow, are symmetry boundaries. The domain size, boundary conditions and computational grid (body-fitted mesh with overall $2.2 \cdot 10^6$ control volumes (CVs), where $0.5 \cdot 10^6$ CVs were used inside the particle) have been taken from the work by Richter & Nikrityuk (Richter and Nikrityuk, 2012). The grid and domain size studies were carried out in the work (Richter and Nikrityuk, 2012). It should be noted that the grid resolution inside the particle is significantly higher than in works (Dixon et al., 2010; Pashchenko and Eremin, 2021). For this reason we did not conduct a separate grid study.

The set of all CFD equations and CFD software validation can be found in the recent work by Pashchenko and Eremin (Pashchenko and Eremin, 2021). In this work the following assumptions were utilized:

- Due to the small velocity magnitudes compared to the local speed of sound, the fluid behaves as a nearly incompressible medium, so the fluid density is treated as an ideal incompressible gas.
- Viscous heating and buoyancy effects are neglected.
- Thermal diffusion is neglected.
- The flow is laminar and steady.
- The particle is porous and its porosity is isotropic, the particle porosity is $\varepsilon = 0.5$.
- The solid and gas phase are in thermal equilibrium inside the porous particle.
- The diffusion coefficient for each species is calculated using the kinetic theory of gases. Inside the porous particle, the diffusion coefficient is computed using Eq. (21).

The effective thermal conductivity of the porous particle, k_{eff} , is computed as the volume average of the fluid conductivity and the solid conductivity:

$$k_p = k_{eff} = \varepsilon k_g + (1 - \varepsilon) k_s \quad (42)$$

where ε is the porosity of the particle, k_g is the gas phase thermal conductivity, k_s is the solid medium thermal conductivity. In this work we used $c_{p,p} = 1000$ J/kg K, $\rho_p = 1947$ kg/m³ and $k_p = 1$ W/mK, see the work (Dixon, 2017). The molecular viscosity, which is calculated using the kinetic theory. This option is available in the

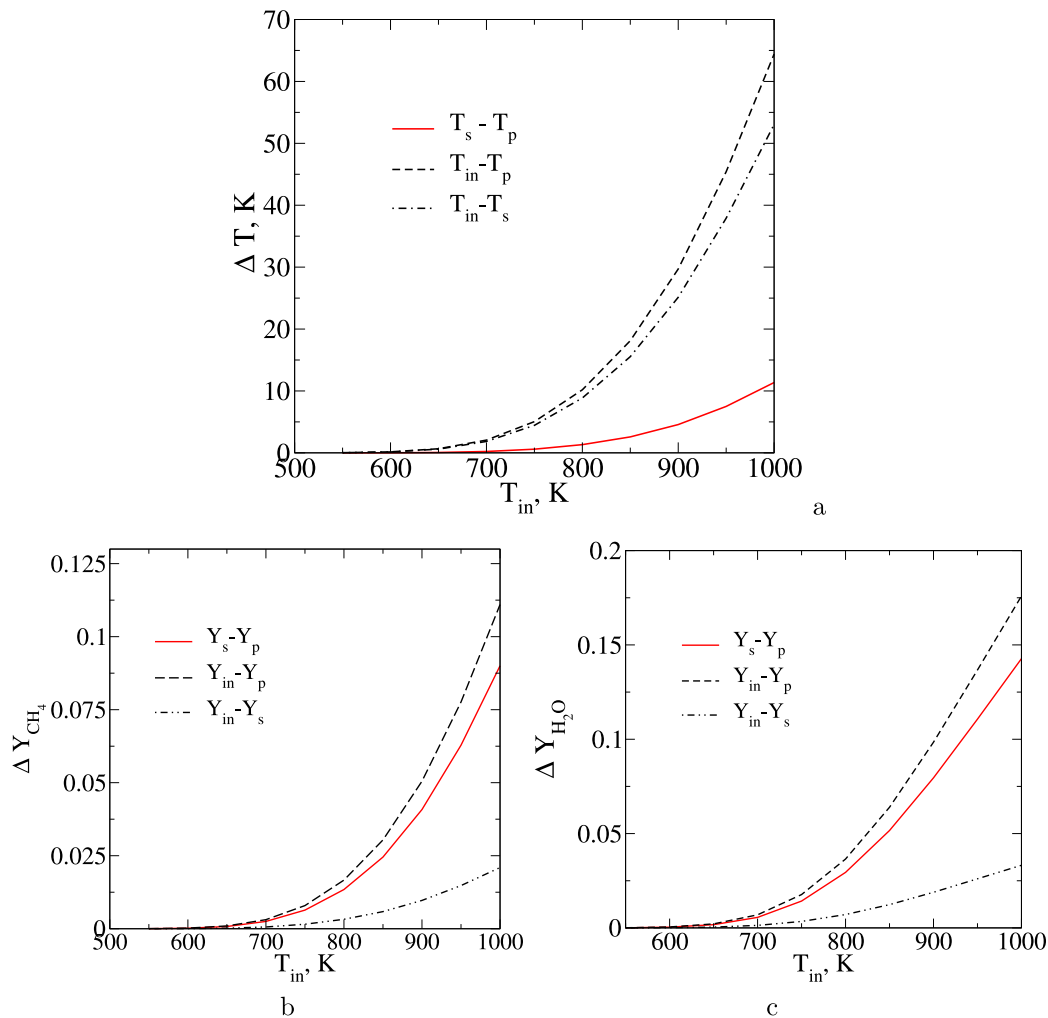


Fig. 7. 0-D model: change in the difference between T_s , T_p and T_{in} (a) and Y_{is} , Y_{ip} and Y_{in} of CH_4 (b) and H_2O (c) depending on the ambient temperature T_{in} for $Re = 100$.

CFD software Ansys-Fluent 19.1 which was used for CFD simulations.

The flow inside the porous particle is simulated using the Darcy-Forchheimer term added to momentum conservation equations:

$$S_i = - \left(\sum_{j=1}^3 \frac{1}{\alpha} \mu v_j + \sum_{j=1}^3 C_2 \frac{1}{2} \rho |v| v_j \right) \quad (43)$$

where S_i is the source term for the i th (x, y , or z) momentum equation; $|v|$ is the magnitude of the velocity; α is the permeability and C_2 is the inertial resistance factor. The choice of the Darcy-Forchheimer term is explained by taking into account inertial effects for flows in the porous particle for $Re > 10$.

The permeability and inertial loss coefficient in each component direction were calculated as follows:

$$\alpha = \frac{D_{pore}^2}{150} \frac{\varepsilon^3}{(1 - \varepsilon)^2} \quad (44)$$

and

$$C_2 = \frac{3.5}{D_{pore}} \frac{(1 - \varepsilon)}{\varepsilon^3} \quad (45)$$

where D_{pore} is the pore size; ε is the porosity of the particle. In this work we used $D_{pore} = \frac{d_p}{100}$, $\varepsilon = 0.5$.

To implement an intrinsic reaction mechanism, the chemical kinetic model described in Section (2.4) was hooked to CFD software Ansys-Fluent 19.1 using UDF 'DEFINE_VR_RATE'. The parameters of the chemical model for the CFD model and for the semi-empirical model are the same. Examples of UDFs for the kinetic mechanism defined by Xu and Froment (Jianguo and Froment, 1989) can be found in the work by Pashchenko and Eremin (Pashchenko and Eremin, 2021). Finally, we emphasize that the CFD software used in this work have been validated against experimental data for steam methane reforming published in the literature, see the work by Pashchenko and Eremin (Pashchenko and Eremin, 2021).

4. Results

Before we compare the predictions obtained using 0-D and 3D CFD models, we want to describe the main findings calculated utilizing the 3D particle-resolved CFD model. Referring to flow past a spherical particle in a steady-state regime, there are several flow features which should be mentioned. First, for $Re = 100$ (considered in this work) the flow past a sphere is steady laminar and axisymmetric. For this Reynolds number, the isothermal flow past a sphere features a so-called recirculating zone (vortex ring), which develops close to the rear stagnation point. This stagnation point is known as the backward stagnation point. A forward stagnation point is located on the forward side of the sphere surface, namely

on the symmetry axis. The forward and backward stagnation points are characterized by the largest and lowest local temperature gradients on the sphere surface (Richter and Nikrityuk, 2012).

Figs. 2 and 3 show 3-D distributions of the temperature and species Y_{CH_4} and Y_{H_2} inside the catalyst particle, respectively. In particular, from Fig. 2 it can be seen that inside the particle the temperature drops rapidly due to endothermic reactions in the porous catalyst. The mass fraction of CH_4 decreases and the mass fraction of H_2 increases in the particle, see Fig. 3. The increase in the ambient temperature T_{in} enhances the drop in the temperature and in Y_{CH_4} . Comparing Fig. 2a and c shows that increase in T_{in} from 600 K to 1000 K causes a significant decrease in the particle temperature such that the temperature in the recirculation zone decreases significantly. It should be noted that the particle surface is not isothermal.

For a qualitative illustration of changes in the species mass fractions and temperature inside the particle, Fig. 4 depicts axial profiles of T , the methane mass fraction, Y_{CH_4} and hydrogen mass fraction, Y_{H_2} along the particle symmetry axis at $z = 0$ and $y = 0$. Analysis of the profiles revealed strong changes in the temperature and species mass fractions inside the particle with increase in the ambient temperature from 800 K to 1000 K. The minimum temperature inside the particle was detected at $x/d_p = 0.11$ from the forward stagnation point of the sphere, $x = 0$. This distance corresponds to the minimum values for the methane mass fraction and maximum values for the hydrogen mass fraction. The CH_4 mass fraction decreases inside the particle between $x/d_p = 0$ and $x/d_p = 0.11$, reaching a minimum at $x/d_p = 0.11$. An increase in the ambient temperature T_{in} leads to an increase in the temperature gradients and mass fractions of all species inside the particle. It should be noted, that T_{min} does not correspond to the particle-averaged temperature T_p used in Eq. (12). T_{min} is the minimum temperature inside the particle.

To estimate the scale of the temperature drop and change in species mass fractions inside the particle, Fig. 5 illustrates the T_{in} -dependence of a change in the temperature difference ($T_{in} - T_{min}$), a change in the relative temperature difference $\frac{T_{in} - T_{min}}{T_{in}}$ and a change in the relative difference of the CH_4 mass fraction between ambient values and minimum values inside the particle. It can be seen that increase in T_{in} leads to a more significant non-linear decrease in the CH_4 mass fraction inside the particle (up to 80%) in comparison to the temperature drop (up to 15%).

Finally, to compare the results of 3D CFD model against the predictions of the 0-D model, we use the volume-averaged quantities such as the particle-averaged temperature and particle-averaged mass fractions of each species, calculated as follows:

$$T_p = \frac{1}{V_p} \int_{V_p} T(x, y, z) dV_p \quad (46)$$

$$Y_{i,p} = \frac{1}{V_p} \int_{V_p} Y_i(x, y, z) dV_p \quad (47)$$

where $V_p = \frac{4}{3} \pi r_p^3$ is the particle volume.

Fig. 6 depicts a comparison of the particle-averaged temperature T_p and particle-averaged species mass fractions of species obtained from 3D CFD simulations and T_p , $Y_{i,p}$ predicted using 0-D model for $Re = 100$. Excellent agreement between the two models can be observed. In particular, the 0-D model precisely predicts all the trends relating to the change in the particle temperature and species mass fractions with the increase in the ambient temperature T_{in} . Namely, an increase in T_{in} from 800 K to 1000 K leads to a significant decrease in the CH_4 and H_2O mass fractions inside the particle and a substantial increase in the H_2 , CO and CO_2 mass fractions. To estimate the role of the particle surface-averaged tem-

perature, T_s and species mass fractions $Y_{s,i}$ in the heat and mass transfer between the bulk flow and particle, Fig. 7 plots the difference between T_s , T_p and T_{in} , and $Y_{i,s}$, $Y_{i,p}$ and $Y_{i,in}$ predicted using the 0-D model for different ambient temperatures and $Re = 100$. Analysis of this figure revealed that, firstly, the particle surface-averaged values of the temperature and mass fractions, T_s , $Y_{s,i}$, deviate substantially from T_p and $Y_{p,i}$, which characterize the particle internal temperature and mass fractions of the species. The difference between the particle surface-averaged mass fractions of species CH_4 and H_2 and of the same particle-averaged quantities, $(Y_s - Y_p)$, is more substantial in comparison to the temperature $(T_s - T_p)$. Finally, it should be noted that $(T_s - T_p) \neq (T_{in} - T_{min})$, because T_{min} is the minimum temperature inside the particle and T_p reflects the particle-averaged temperature. It can be seen comparing Figs. 7a and 5.

5. Conclusions

This work presents the development and validations of a new 0-D model for the steam reforming of methane inside a spherical catalyst particle ($Ni/\alpha-Al_2O_3$) placed in a hot methane/steam atmosphere. The 0-D model includes six gaseous chemical species (CH_4 , CO_2 , CO, H_2O , H_2 , N_2) using a global mechanism of LHHW kinetics taken from the literature. This new model predicts the particle surface-averaged temperature and species mass fractions, and intraparticle values of the temperature and species mass fractions. Introducing surface- and interparticle-relevant variables enables calculations of the particle internal heat and mass transfer coupled with the heat and mass transfer on the catalyst surface influenced by convection and diffusion around the particle. The 0-D model has been validated against a comprehensive 3D-CFD-based model resolving the issues of bulk flow, thermal and species boundary layers around the particle and convection-diffusion processes inside the porous catalyst particle. Excellent agreement was achieved between the results of the new 0-D model and the 3D particle-resolved CFD model. The results of CFD simulations revealed a considerable change in the temperature and species mass fractions inside the particle as the ambient temperature increased from 800 K to 1000 K. The minimum temperature inside the particle was detected at $x/d_p = 0.11$ from the forward stagnation point of the sphere. This distance corresponds to the minimum values for the methane mass fraction and maximum values for the hydrogen mass fraction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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